# LG100N10AG

N-Channel Enhancement Mode MOSFET

## FEATURES

- Best-in-Class FOM
- Low Gate Charge
- High Current Capability

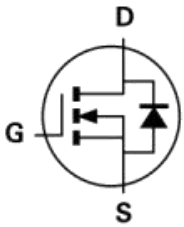
## MAIN CHARACTERISTICS

| | |
|---|---|
| $I_D$ | 100A |
| $V_{DSS}$ | 100V |
| $R_{DSON}$-typ (@$V_{GS}$=10V) | 5.3mΩ |

## APPLICATIONS

- Power Management in Telecom.,Industrial Automation
- Motor Driving in Power Tool,E-vehicle,Robotics
- Current Switching in DC/DC&AC/DC(SR) Sub-system

- Case: Molded plastic
- Mounting Position: Any
- Molded Plastic: UL Flammability Classification Rating 94V-0
- Lead free in compliance with EU RoHS 2011/65/ EU directive
- Solder bath temperature 275℃ maximum,10s per JESD 22-B106




PDFN5x6

## Maximum Ratings at Tc=25°C unless otherwise specified

| Characteristics | Symbol | Value | Unit |
|---|---|---|---|
| Drain-Source Voltage | $V_{DS}$ | 100 | V |
| Gate-Source Voltage | $V_{GS}$ | ±20 | V |
| Continue Drain Current | $I_D$ | 100 | A |
| Pulsed Drain Current (Note1) | $I_{DM}$ | 360 | A |
| Power Dissipation | $P_D$ | 96 | W |
| Single Pulse Avalanche Energy (Note1) | $E_{AS}$ | 216 | mJ |
| Operating Temperature Range | $T_J$ | 150 | °C |
| Storage Temperature Range | $T_{STG}$ | -55 to +150 | °C |
| Thermal Resistance, Junction to Case | $R_{\theta JC}$ | 1.7 | °C/W |
| Thermal Resistance, Junction to Ambient | $R_{\theta JA}$ | 50 | °C/W |

Note1:Pulse test: 300 μs pulse width, 2 % duty cycle

## Electrical Characteristics at Tc=25°C unless otherwise specified

| Characteristics | Test Condition | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|
| Drain-Source Breakdown Voltage | $V_{GS}$ = 0 V,$I_D$ = 250 μA | $BV_{DSS}$ | 100 | - | - | V |
| Drain-Source Leakage Current | $V_{DS}$ = 100 V, $V_{GS}$ = 0 V | $I_{DSS}$ | - | - | 1 | μA |
| Gate Leakage Current | $V_{GS}$ = ± 20 V, $V_{DS}$ = 0 V | $I_{GSS}$ | - | - | ±100 | nA |
| Gate-Source Threshold Voltage | $V_{DS}$ = $V_{GS}$ , $I_D$ = 250 μA | $V_{GS(th)}$ | 1.2 | - | 2.5 | V |
| Drain-Source On-State Resistance | $V_{GS}$ = 10 V, $I_D$ = 20A | $R_{DS(on)}$ | - | 5.3 | 6.4 | mΩ |
| | $V_{GS}$ = 4.5 V, $I_D$ = 15A | $R_{DS(on)}$ | - | 7.5 | 9.4 | mΩ |
| Input Capacitance | VDS=50V , VGS=0V , f=1MHz | $C_{iss}$ | - | 2420 | - | pF |
| Output Capacitance | | $C_{oss}$ | - | 900 | - | pF |
| Reverse Transfer Capacitance | | $C_{rss}$ | - | 35 | - | pF |
| Turn-on Delay Time(Note2) | VDD=50V , VGS=10V , RG=3Ω , ID=20 A | $t_{d(ON)}$ | - | 11.2 | - | ns |
| Rise Time(Note2) | | $t_r$ | - | 23 | - | ns |
| Turn-Off Delay Time(Note2) | | $t_{d(OFF)}$ | - | 38 | - | ns |
| Fall Time(Note2) | | $t_f$ | - | 15.3 | - | ns |
| Total Gate Charge(Note2) | VDS=50V , VGS=10V , ID=20A | $Q_G$ | - | 43 | - | nC |
| Gate to Source Charge(Note2) | | $Q_{GS}$ | - | 9.4 | - | nC |
| Gate to Drain Charge(Note2) | | $Q_{GD}$ | - | 10.3 | - | nC |

## Source-Drain Diode Characteristics at Ta=25°C unless otherwise specified

| Characteristics | Test Condition | Symbol | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| Maximun Body-Diode Continuous Current | | $I_S$ | - | - | 100 | A |
| Drain-Source Diode Forward Voltage | VGS=0V , IS=20A , TJ=25° | $V_{SD}$ | - | - | 1.2 | V |
| Reverse Recovery Time(Note2) | $T_J$ = 25℃, $I_F$= 20A di / dt = 100 A/μs | trr | - | 42 | - | ns |
| Reverse Recovery Charge(Note2) | | Qrr | - | 46 | - | nC |

Note2:Pulse test: 300 μs pulse width, 2 % duty cycle

## RATINGS AND CHARACTERISTIC CURVES

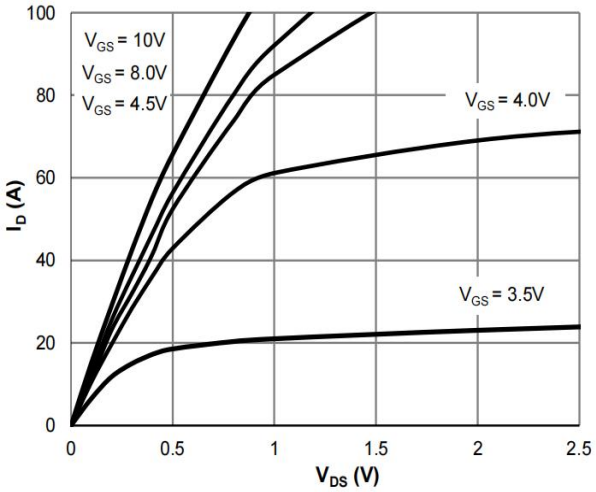



**Figure 1: Saturation Characteristics**

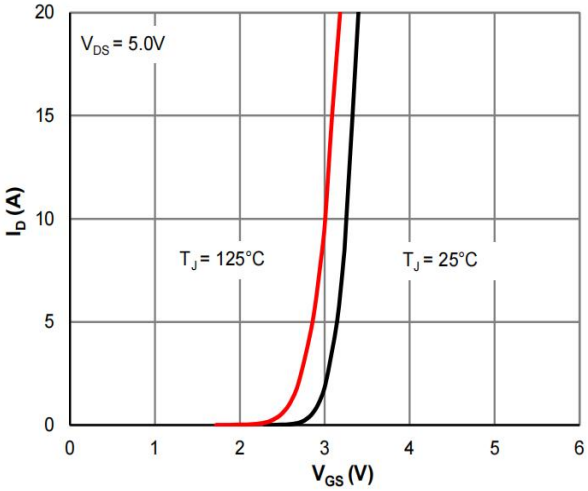



**Figure 2: Transfer Characteristics**

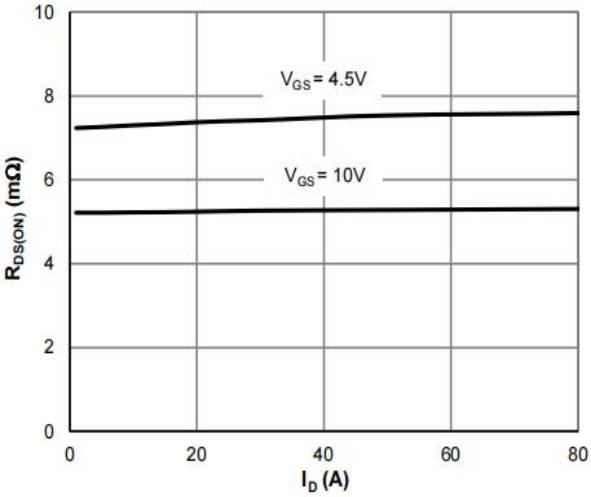



**Figure 3: $R_{DS(ON)}$ vs. Drain Current**

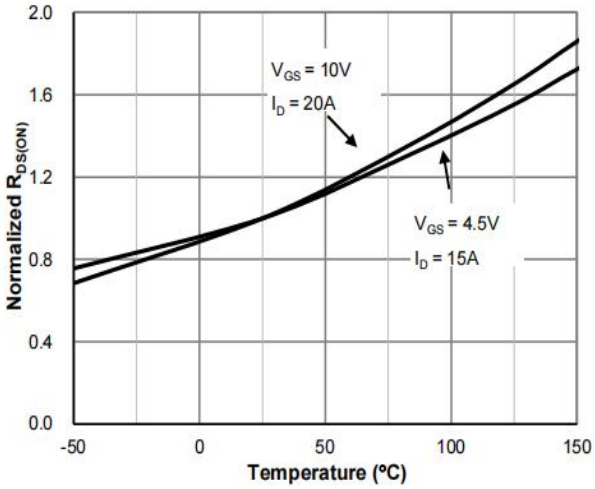



**Figure 4: $R_{DS(ON)}$ vs. Junction Temperature**

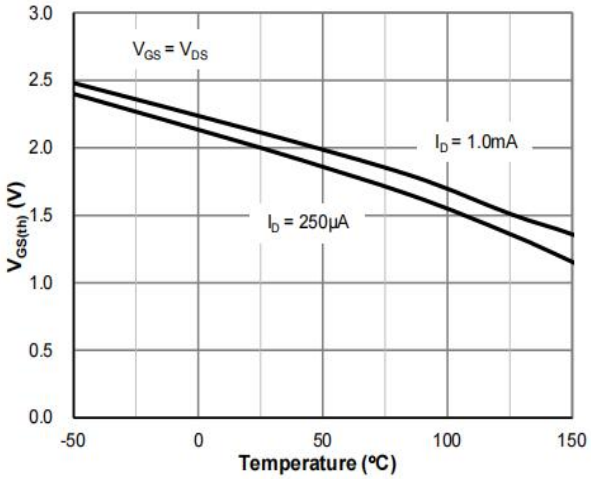



**Figure 5: $V_{GS(th)}$ vs. Junction Temperature**

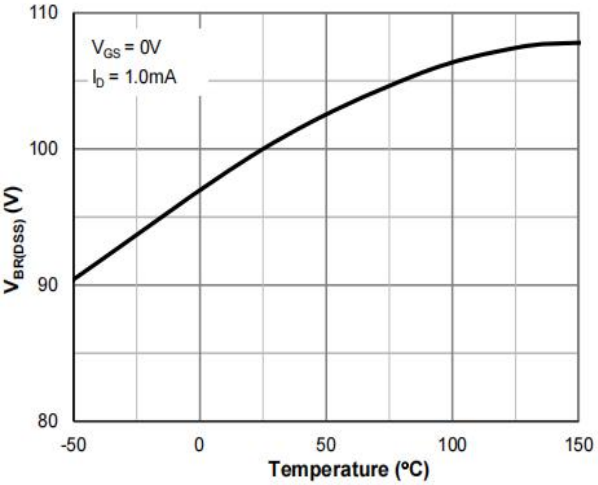



**Figure 6: $V_{BR(DSS)}$ vs. Junction Temperature**

## RATINGS AND CHARACTERISTIC CURVES

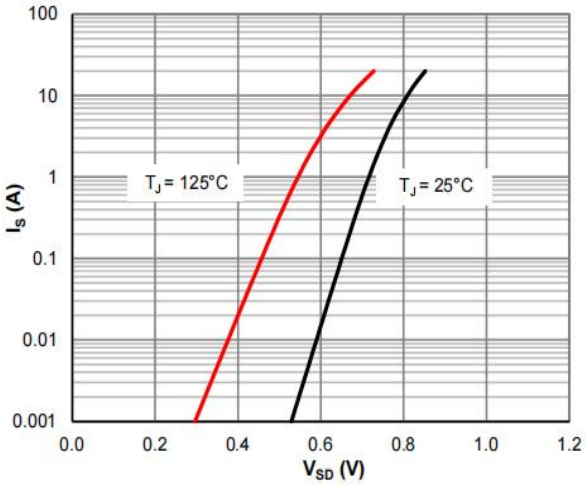



Figure 7: Body-Diode Characteristics

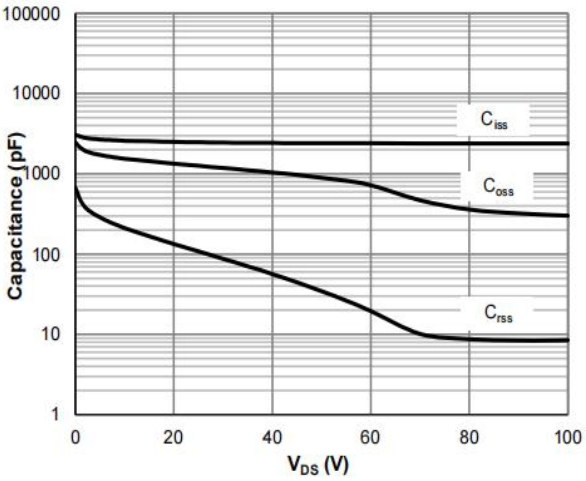



Figure 8: Capacitance Characteristics

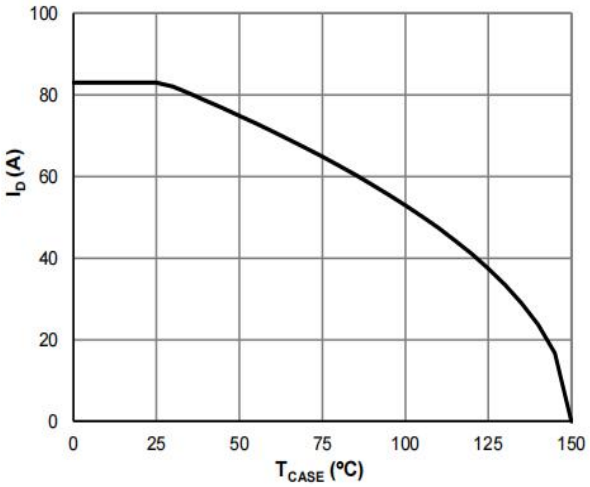



Figure 9: Current De-rating

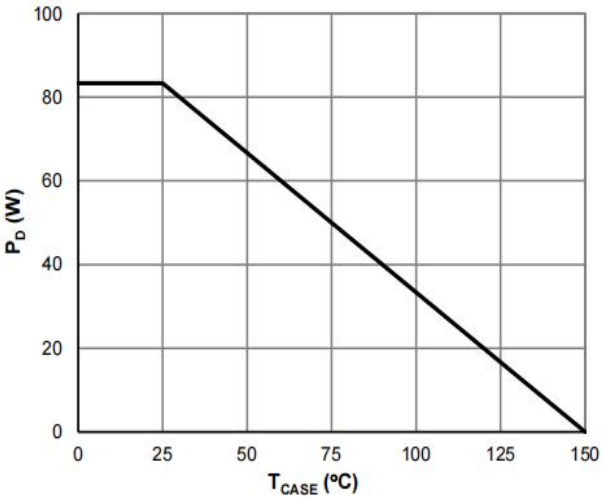



Figure 10: Power De-rating

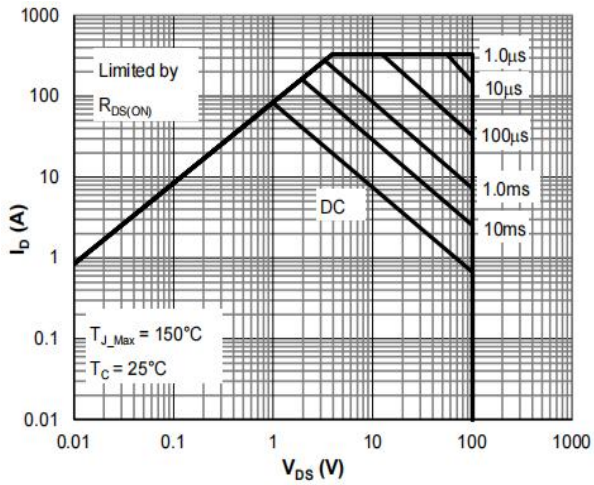



Figure 11: Maximum Safe Operating Area

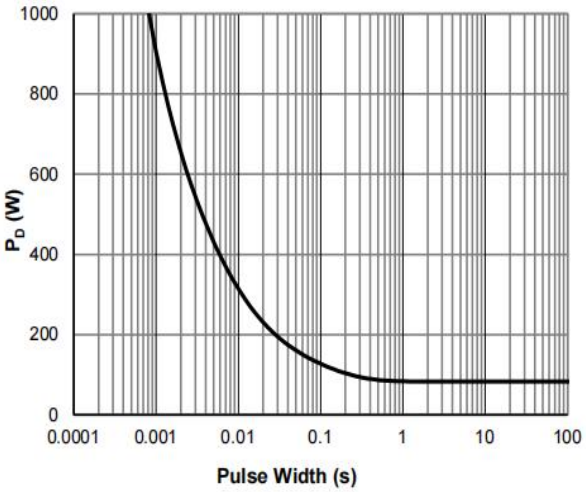
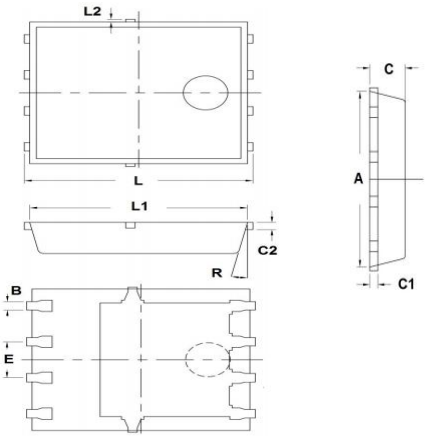



Figure 12: Single Pulse Power Rating, Junction-to-Case

## Package Outline Dimensions millimeters




PDFN5*6

| Dim. | Min. | Max. |
|---|---|---|
| A | 4.8 | 5.2 |
| B | 0.25 | 0.35 |
| C | 1 | 1.2 |
| C1 | Typ0.254 | |
| C2 | Typ0.254 | |
| E | Typ1.27 | |
| L | 6 | 6.3 |
| L1 | 5.7 | 6 |
| L2 | Max0.2 | |
| R | Typ13° | |
| All Dimensions in millimeter | | |